

disk, graphics adapter, etc. The kernel component of KVM is included in mainline Linux, as of 2.6.20. The user space component of KVM is included in mainline QEMU, as of 1.3. Blogs by people active in KVM-related virtualisation development are syndicated at <http://planet.virt-tools.org/>.

Xen hypervisor is an open source Type 1 or bare metal hypervisor, which makes it possible to run many instances of an operating system or different operating systems in parallel on a single machine (or host). The Xen Project hypervisor is the only Type 1 hypervisor that is available as open source. It is used as the basis for a number of different commercial and open source applications, such as server virtualisation, Infrastructure as a Service (IaaS), desktop virtualisation, security applications, and embedded and hardware appliances. The Xen Project hypervisor is powering the largest clouds in production today.

Container based virtualisation: Container based virtualisation, also known as operating system-level virtualisation, enables multiple isolated executions within a single operating system kernel. It has the best possible performance and density, while featuring dynamic resource management. The isolated virtual execution environment provided by this type of virtualisation is called a container and can be viewed as a traced group of processes.

Containers have also sparked an interest in micro service architecture, a design pattern for developing applications in which complex applications are broken down into smaller pieces that work together. Each component is developed separately, and the application is then simply the sum of its constituent components. Each piece, or service, can reside inside a container, and can be scaled independently of the rest of the application as the need arises.

Undoubtedly, one of the biggest reasons for the recent interest in container technology has been the Docker open source project, a command line tool that makes creating and working with containers easy for developers and sys admins alike, similar to the way Vagrant makes it easier for developers to explore virtual machines easily.

Docker is a command line tool for programmatically defining the contents of a Linux container in code, which can

then be versioned, reproduced, shared and modified easily just as if it were the source code to a program.

Cloud platforms

Cloud Foundry is the world's leading open source Platform-as-a-Service (PaaS) for continuous innovation

The Cloud Foundry platform is available from either the Cloud Foundry Foundation as open source software, or from multiple commercial providers as a product or a service.

Cloud Foundry is open source software and, hence, available to anyone. Deploying it involves interfacing with the underlying infrastructure using the Cloud Foundry BOSH (bosh outer shell) deployment scripting language, another open source tool from Pivotal. The Baidu website is implemented on Cloud Foundry.

Orchestration tools

Based on the cloud platforms adopted, given below are some of the top orchestrators.

Cloud Foundry's Diego: Diego is a container management system that combines a scheduler, runner and health manager. It is a rewrite of the Cloud Foundry runtime.

Docker Swarm: Docker Swarm provides native clustering functionality for Docker containers, which lets you turn a group of Docker engines into a single, virtual Docker engine.

Kubernetes: Kubernetes is an orchestration system for Docker containers. It handles scheduling and manages workloads based on user-defined parameters.

Mesosphere Marathon: Marathon is a container orchestration framework for Apache Mesos that is designed to launch long-running applications. It offers key features for running applications in a clustered environment.

Big Data applications

Apache Cassandra is a free and open source distributed database management system designed to handle large amounts of data across many commodity servers, providing high availability with no single point of failure. Cassandra offers robust support for clusters spanning multiple data centres, with asynchronous masterless replication allowing low latency operations for all clients.

Hadoop is an open source, Java-based programming framework that supports the processing and storage of extremely large data sets in a distributed computing environment. It is part of the Apache project, sponsored by the Apache Software Foundation

Apache Spark is an open source cluster computing framework. Originally developed at the University of California, Berkeley's AMPLab, the Spark codebase was later donated to the Apache Software Foundation, which has maintained it since. Spark provides an interface for programming entire clusters with implicit data parallelism and fault-tolerance. It provides programmers with an application programming interface centred on a data structure called the resilient distributed dataset (RDD),

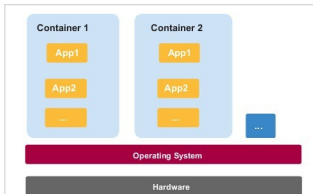


Figure 7: Container based virtualisation